UNIT I

SHORT ANSWER QUESTIONS

1. Define Principle of superposition.
2. What are coherent sources?
3. What is interference of light waves?
4. Give the conditions for producing good interference fringes. (2M or 5M)
5. Explain the formation of colours in thin films.(2M or 5M)
6. What are the differences between interference and diffraction? (2M or 5M)
7. What is diffraction? Write the condition to get diffraction of light.
8. Distinguish between Fresnel and Fraunhofer diffraction.(2M or 5M)
9. What is meant by diffraction grating?
10. Explain resolving power and dispersive power of a grating.
11. What is polarization?
12. What are the different types of polarized light?
13. State Brewster's law.

LONG ANSWER QUESTIONS:

CHAPTER-I

1. With a ray diagram discuss the phenomenon of interference in thin films and obtain the conditions for maxima and minima.
2. Derive an expression for the diameters of the dark and bright rings formed in Newton's rings arrangement.
3. Derive an expression for the wavelength of the source in Newton's rings arrangement.
4. Obtain an expression for refractive index of any liquid using Newton's rings arrangement.

CHAPTER-II

5. Describe the Fraunhofer diffraction pattern due to single slit and obtain the condition for primary and secondary maxima, width of the central maxima.
6. Explain Fraunhofer diffraction due to double slit.
7. Explain Fraunhofer diffraction due to multiple slits.
8. Explain Rayleigh's criterion for resolving power and hence discuss the resolving power of a grating.

CHAPTER-III

9. State Brewster's law.Show that at polarizing angle the reflected and refracted rays are perpendicular to each other.
10. Explain the phenomenon of double refraction. How can it be used to produce a plane polarized light.
11. Mention the properties of ordinary and extraordinaory rays.
12. Describe the construction and working of Nicol prism and explain how it acts as a polarizer and analyser.
13. Derive an expression for thickness of quarter wave plate and half wave plate.
14. Problems.

UNIT II

SHORT ANSWER QUESTIONS

1. What are dielectric materials? Give examples.
2. What are non polar and polar dielectric materials?
3. What is polarization and mention the different types of polarization. (2M or 5M)
4. Define the following terms: polarization vector, electric displacement, relative permittivity (or) dielectric constant, dielectric susceptibility and obtain the relation between them.
5. Discuss dielectric loss.
6. Mention the applications of dielectric materials.
7. Define the following terms: Magnetic field intensity, Magnetic induction, Magnetic permeability Magnetic susceptibility and obtain the relation between magnetic susceptibility and permeability
8. Derive the relation $B=(H+I)\mu_0$
9. What is Bohr magneton? Write its expression.
10. Discuss the applications of magnetic materials.

LONG ANSWER QUESTIONS

CHAPTER-I

1. Derive an expression for electronic polarizability in terms of atomic radius.
2. What is ionic polarization? Derive an expression for ionic polarizability.
3. Explain the concept of local field and derive an expression for it.
4. Discuss the variation of polarizability of a dielectric as a function of the frequency.
5. Derive Claussius-Mosotti relation in dielectrics.

CHAPTER--II

6. Explain the origin of magnetic moment in an atom and obtain the expression for magnetic moment due to the orbital motion of an electron and its spin.(or)

   What is Bohr Magneton? Derive its expression.
7. Explain the classification of magnetic materials in atomic point of view (dia, para, ferro antiferro, ferri magnetic materials)
8. Explain Weiss theory (domain theory) of ferromagnetism.
9. Explain hysteresis (B-H) curve in ferromagnetic materials.
10. Distinguish between soft and hard magnetic materials.

UNIT III

SHORT ANSWER QUESTIONS

1. Explain the terms: space lattice, basis, unit cell, lattice parameters.(2Mor5M)
2. What is primitive cell?
3. Define :Bravais lattice, atomic radius, coordination number, packing fraction.
4. What are Bravais lattices.?
5. List the steps to derive the Miller indices of a plan.What are the important features of Miller indices?
6. Sketching the planes.

LONG ANSWER QUESTIONS

CHAPTER--I

1. Describe with diagrams the seven crystal systems and the Bravais lattices.
2. Show that FCC is the most closely packed structure by working out the packing factors.
3. Derive an expression for interplanar distance of separation of a cubic crystal.

CHAPTER-II

4. State and explain Bragg's law of diffraction.
5. Explain Bragg's X-Ray diffractometer.
6. Explain Laue method for studying crystal structure.
7. Explain powder method or Debye Scherrer method for studying crystal structure.
8. Problems.

UNIT IV

SHORT ANSWER QUESTIONS

1. Explain de Broglie's hypothesis and derive the expressions for de Broglie's wavelength.
2. What are matter waves? Write the properties of matter waves.(2Mor 5M)

3. Discuss the merits and drawbacks of Classical free electron theory. (2M or 5M)
4. What are the salient features, merits and demerits of Quantum free electron theory? (2M or 5M)
5. Problems on debroglie's wave length.
6. Define Fermi energy.

LONG ANSWER QUESTIONS

CHAPTER-I

1. State Heisenberg's uncertainity principle.
2. Explain the physical significance of wave function and write the conditions for a physically acceptable wave function (limitations of wave function)
3. Derive Schrodinger's time independent wave equation.
4. Derive Schrodinger's time dependent wave equation.
5. Discuss the behavior of an electron moving in a one dimensional potential well and obtain an expression for its energy. (or)

   Apply Schrodinger's timeindependent wave equation to a particle in a one dimensional potential well and derive an expression for its energy

CHAPTER--II

6. Explain Fermi Dirac distribution and its temperature dependence. Illustrate graphically.
7. Derive an expression for electrical conductivity on the basis of quantum free electron theory.
8. Derive an expression for density of states.

UNIT V

SHORT ANSWER QUESTIONS

1. What are intrinsic and extrinsics semiconductors?
2. Give examples of trivalent and pentavalent impurities.
3. Distinguish between conductors, semiconductors and insulators.(2M or 5M)
4. Write the differences between intrinsic and extrinsic semiconductors. (2M or 5M)
5. Write the expressions for density of charge carriers in intrinsic and extrinsic semiconductors.
6. Drift and diffusion current expressions,
7. Write the expressions for Einstein's relations.

LONG ANSWER QUESTIONS

1. What is an intrinsic semiconductor? Derive an expression for carrier concentration of electrons in the conduction band of an intrinsic semiconductor.
2. Derive an expression for carrier concentration of holes in the valence band of an intrinsic semiconductor.

3. Derive an expression for carrier concentration of an intrinsic semiconductor.
4. Show that Fermi energy level lies in the middle of conduction energy band and valence energy band in an intrinsic semiconductor.
5. Derive an expression for conductivity of an intrinsic semiconductor.
6. Derive carrier concentration of electrons and holes in n-type and p-type extrinsic semiconductors.
7. Discuss the variation of Fermi energy level with temperature and impurity concentration in extrinsic semiconductors.
8. What do you understand by drift and diffusion currents in semiconductor? Derive their Expressions .
9. Derive Einstein's relations.
10. State and explain Hall effect. What is Hall coefficient ? What are its applications.
11. Problems.

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